

Prior Specification

Every prior with its epistemic label and source

Labels: LITERATURE traceable to a citation; STRUCTURAL required for well-posedness; WEAK weakly informative regularisation justified by prior predictive checking; DISCLOSURE deliberately non-informative because the parameter is known to be unidentified.

Ebola

Parameter	Value	Label	Source
structure	seird_incidence	STRUCTURAL	LITERATURE: Legrand J et al. (2007) Epidemics 21(3):183-197 -- SEIHFR structure for Ebola with isolation; we use the reduced SEIRD form with isolation folded into the removal rate. Observable is daily onset incidence (Khan AS et al. (1999) J Infect Dis 179:S76-86).
sigma	0.1111 /day	LITERATURE	LITERATURE: WHO Ebola fact sheet (2-21 d incubation); Legrand et al. (2007) 10 d mean
removal	0.1667 /day	LITERATURE	LITERATURE: Legrand et al. (2007) Table 2, $\gamma_I = 0.167 \text{ d}^{-1}$; WHO 4-9 d symptomatic period. Post-intervention isolation is absorbed by the GRW on $\log \beta(t)$ rather than by a time-varying δ .
N	1000	LITERATURE	STRUCTURAL: Kikwit General Hospital catchment. The absolute value is weakly identified; $\log_intercept_e$ absorbs the scale. Sensitivity over N in {500, 1000, 2000} is prespecified.
nb_alpha_scale	HalfNormal(3.0)	WEAK	WEAK: HalfNormal(3.0), calibrated to pre-fit Var/Mean = 2.60 (static by design; see docs/PRIOR_CALIBRATION.md sec 4)
log_intercept	Normal(0.0, 0.5)	WEAK	WEAK: Normal(0, 0.5) on the log ascertainment ratio; tightened from v5 Normal(-1.8, 1.5) because the v6 trajectory is R_0 -dependent and a wide intercept re-absorbs the scale that identifies R_0
sigma_rw_scale	HalfNormal(0.02)	WEAK	WEAK: prior predictive calibrated (scripts/03); p95 of the implied $\beta(t)$ range stays under 3x. Gelman et al. (2020) BDA3 sec 6.4; Gabry et al. (2019) JRSSA 182:389

Measles

Parameter	Value	Label	Source
structure	seird_incidence	STRUCTURAL	LITERATURE: Neal PJ & Roberts GO (2004) Biostatistics 5(2):249-261 model this outbreak as an SEIR process on the 188-child cohort. Meyer S, Held L & Hoehle M (2017) J Stat Softw 77(11) distribute the linelist. Observable is daily rash-onset incidence.
sigma	0.0909 /day	LITERATURE	LITERATURE: measles incubation 10-14 days to rash onset (CDC Pink Book, ch. 13); Neal & Roberts (2004) use a latent period of this order for this dataset. $1/11 \text{ d}^{-1}$.
removal	0.1250 /day	LITERATURE	LITERATURE: measles infectious from ~4 days before to ~4 days after rash onset (CDC Pink Book, ch. 13), i.e. ~8 days. $1/8 \text{ d}^{-1}$.
N	188	LITERATURE	LITERATURE: complete village child cohort, 188 children (Pfeilsticker 1863; Oesterle 1992). Known exactly -- every child in Hagelloch is in the linelist, so N carries no assumption.
nb_alpha_scale	HalfNormal(3.0)	WEAK	WEAK: HalfNormal(3.0), the same weakly informative scale used for the Ebola incidence arm. Static by design -- see docs/PRIOR_CALIBRATION.md and PARAMETER_CHANGE_AUDIT.md sec 10.
log_intercept	Normal(0.0, 0.0)	WEAK	STRUCTURAL: fixed at 0 (ascertainment = 1). Not estimated.
sigma_rw_scale	HalfNormal(0.05)	WEAK	WEAK: prior predictive calibrated (scripts/03). Only applies when --grw-measles is passed; the default is no GRW, subject to the scripts/02 decision rule.

Influenza

Parameter	Value	Label	Source
structure	seiqr_prevalence	STRUCTURAL	LITERATURE: Avilov KK, Li Q, Lin L, Demirhan H, Stone L, He D (2024). The 1978 English boarding school influenza outbreak: where the classic SEIR model fails. J R Soc Interface 21(220):20240394 -- documents that confined-to-bed (B) must be modelled as a compartment distinct from infectious (I), and that SEIR models treating them as identical significantly overestimate the outbreak timespan.
sigma	0.6667 /day	LITERATURE	LITERATURE: influenza A/H1N1 latent period 1-2 d; Ferguson NM et al. (2003) Science 300:1966
removal	0.5000 /day	LITERATURE	LITERATURE: boys 'remained infectious, i.e. capable of transmitting the disease, for about 2 days' before confinement (BMJ 1978;1(6112):587 and standard treatments of this dataset). $\delta = 0.5 d^{-1}$.
N	763	LITERATURE	LITERATURE: 763 boys at risk (BMJ 1978;1(6112):587); known exactly, no assumption required
nb_alpha_scale	HalfNormal(200.0)	DISCLOSURE	DISCLOSURE: HalfNormal(200). Posterior/prior-mean ratio was 1.27 at HalfNormal(50) and 1.13 at HalfNormal(200) -- the likelihood is flat in this parameter (Gelman et al. (2020) BDA3 sec 13.3). Reported as unidentified, not as an estimate.
log_intercept	Normal(0.0, 0.0)	WEAK	STRUCTURAL: fixed at 0 (ascertainment = 1). Not estimated.
sigma_rw_scale	HalfNormal(0.1)	WEAK	WEAK: prior predictive calibrated (scripts/03); median implied $\beta(t)$ range 1.32x, p95 2.73x, ceiling 3x. Gelman et al. (2020) BDA3 sec 6.4; Gabry et al. (2019) JRSSA 182:389. NOTE: the Influenza GRW is DISABLED by default -- 14 innovations against 14 observations is a saturated latent process. This scale applies only when --grw-influenza is passed for sensitivity.
gamma_q	0.2000 /day	LITERATURE	LITERATURE: 'the average duration of illness was 5 days' (BMJ 1978;1(6112):587)
t_seed	11.0 days	LITERATURE	LITERATURE + PROFILED: 'The index case was infected by 1978-01-10, and had febrile illness from 1978-01-15 to 1978-01-18' (RECON outbreaks, influenza_england_1978_school). The series begins 1978-01-22, giving a documented 8-12 day unobserved lead-in. Within that documented window $t_{\text{seed}}=11$ minimises the Poisson deviance of the constant-beta fit (scripts/02). Sensitivity over {7, 9, 11, 13} is prespecified. NOTE: t_{seed} is fixed, not estimated -- with 14 observations, jointly identifying t_{seed} and R_0 reintroduces the non-identifiability that v6 exists to remove.

Norovirus

Parameter	Value	Label	Source
structure	final_size	STRUCTURAL	LITERATURE: Kermack WO & McKendrick AG (1927) Proc Roy Soc A 115(772):700-721 final size relation; O'Neill PD & Marks PJ (2005) Stat Med 24(13):2011-2024 for this outbreak.
sigma	0.8333 /day	LITERATURE	LITERATURE: CDC norovirus 12-48 h incubation; Heijne JCM et al. (2012) Epidemics 4(4):164-171 mean 1.2 d
removal	0.5000 /day	LITERATURE	LITERATURE: CDC norovirus 1-3 d illness; Heijne et al. (2012) mean 2 d
N	492	LITERATURE	LITERATURE: complete school roll, 492 students (O'Neill & Marks 2005)
nb_alpha_scale	HalfNormal(1.0)	STRUCTURAL	STRUCTURAL: unused -- Norovirus uses a Bernoulli likelihood
log_intercept	Normal(0.0, 1.0)	WEAK	STRUCTURAL: unused -- attack rate enters directly through the K-M relation
sigma_rw_scale	HalfNormal(0.0)	STRUCTURAL	STRUCTURAL: no GRW -- the data are cross-sectional, with no time index for $\beta(t)$ to vary over

Shared

Parameter	Value	Label	Source
hyper_log_R0_mu	Normal(0.5, 1.0)	WEAK	WEAK: Gelman et al. (2020) BDA3 Ch. 5; Gelman (2006) Bayesian Analysis 1(3):515
hyper_log_R0_sd	HalfNormal(0.5)	WEAK	WEAK: Gelman et al. (2020) BDA3 Ch. 5; Gelman (2006) Bayesian Analysis 1(3):515